

Lung cancer prediction using machine learning and advanced imaging techniques

<https://pmc.ncbi.nlm.nih.gov/articles/PMC6037965/>

- Kadir T, Gleeson F. Lung cancer prediction using machine learning and advanced imaging techniques[J]. Translational lung cancer research, 2018, 7(3): 304.

这项研究发现传统的LDCT方法对于肺癌的检测假阳性概率很高，引入计算机辅助诊断（CADx）辅助医生从而降低误诊率。

Lung Cancer Prediction using Machine Learning: A Comprehensive Approach

<https://ieeexplore.ieee.org/abstract/document/9074947>

- Raoof S S, Jabbar M A, Fathima S A. Lung Cancer prediction using machine learning: A comprehensive approach[C]//2020 2nd International conference on innovative mechanisms for industry applications (ICIMIA). IEEE, 2020: 108-115.

综述了各种类型的机器学习算法（ML），如朴素贝叶斯、支持向量机（SVM）、逻辑回归、人工神经网络（ANN）等模型对于肺癌预测的优缺点

A Risk Model for Prediction of Lung Cancer

<https://academic.oup.com/jnci/article-abstract/99/9/715/2544273>

- Spitz M R, Hong W K, Amos C I, et al. A risk model for prediction of lung cancer[J]. Journal of the National Cancer Institute, 2007, 99(9): 715-726.

探究了环境烟草烟雾、癌症家族史、粉尘暴露、既往呼吸系统疾病和吸烟史变量对于肺癌的影响并构建了一个多变量回归模型

Diagnosis of Lung Cancer Prediction System Using Data Mining Classification Techniques

[Link](#)

- Krishnaiah V, Narsimha G, Chandra N S. Diagnosis of lung cancer prediction system using data mining classification techniques[J]. International Journal of Computer Science and Information Technologies, 2013, 4(1): 39-45.

使用了一个对朴素贝叶斯不稳定概率的扩展模型ODANB和NCC2完成了一个通过年龄，性别，抽搐，呼吸急促等因素帮助对肺癌进行早期发现和正确诊断的模型

Lung cancer prediction by Deep Learning to identify benign lung nodules

<https://www.sciencedirect.com/science/article/pii/S0169500221000453>

- Heuvelmans M A, van Ooijen P M A, Ather S, et al. Lung cancer prediction by Deep Learning to identify benign lung nodules[J]. Lung cancer, 2021, 154: 1-4.

通过深度学习基于ct扫描结果预测恶性肿瘤

Risk Prediction Models for Lung Cancer: A Systematic Review

<https://www.sciencedirect.com/science/article/abs/pii/S1525730415002715>

- Gray E P, Teare M D, Stevens J, et al. Risk prediction models for lung cancer: a systematic review[J]. Clinical lung cancer, 2016, 17(2): 95-106.

对比了不同模型对于已有癌症的风险的预测

Lung cancer risk prediction: A tool for early detection

<https://onlinelibrary.wiley.com/doi/abs/10.1002/ijc.22331>

- Cassidy A, Duffy S W, Myles J P, et al. Lung cancer risk prediction: a tool for early detection[J]. International journal of cancer, 2007, 120(1): 1-6.

发现了年龄和吸烟的影响之外，现有相对较少的模型来预测肺癌风险。因此这篇研究回顾了已有的不同癌症预测模型，并引入了职业暴露、个人病史和癌症家族史等新的因素到模型中。

Lung cancer prediction model using ensemble learning techniques and a systematic review analysis

<https://ieeexplore.ieee.org/abstract/document/9817326>

- Mamun M, Farjana A, Al Mamun M, et al. Lung cancer prediction model using ensemble learning techniques and a systematic review analysis[C]//2022 IEEE World AI IoT Congress (AllIoT). IEEE, 2022: 187-193.

使用了年龄、吸烟、黄手指、焦虑、同伴压力、慢性疾病、疲劳、过敏、喘息、酒精、咳嗽、呼吸急促、吞咽困难和胸痛这些因素训练预测模型并发现XGBoost模型性能明显优于其他技术。

accuracy of 94.42 %, precision of 95.66%, recall of 94.46%, and AUC of 98.14%, respectively.

Lung cancer Prediction and Classification based on Correlation Selection method Using Machine Learning Techniques

<https://journal.qubahan.com/index.php/qaj/article/view/58>

- Abdullah D M, Abdulazeez A M, Sallow A B. Lung cancer prediction and classification based on correlation selection method using machine learning techniques[J]. Qubahan Academic Journal, 2021, 1(2): 141-149.

这个研究检验支持向量机（SVM）、K最近邻（KNN）和卷积神经网络（CNN）这三种分类器的准确率，并发现SVM的效果最好，为95.56%，其次是CNN的CNN为92.11%，KNN为88.40%。

Optimizing Lung Cancer Prediction Models: A Hybrid Methodology Using GWO and Random Forest

https://link.springer.com/chapter/10.1007/978-3-031-82516-3_3

- Tirumanadham N S K M K, Priyadarshini V, Praveen S P, et al. Optimizing lung cancer prediction models: A hybrid methodology using GWO and Random Forest[M]//Enabling Person-Centric Healthcare Using Ambient Assistive Technology, Volume 2: Personalized and Patient-Centric Healthcare Services in AAT. Cham: Springer Nature Switzerland, 2025: 59-77.

这项研究对准确的机器学习算法、随机森林和灰狼优化（GWO）用于确定肺癌结果的性能进行比较分析。在性能方面，随机森林分类器的准确率为97.21%，准确率和召回率分别为95.02%和92.11%。这些统计数据支持有关模型可以准确预测的肺癌类型的证据，这对临床诊断很重要。

Comparative analysis of machine learning techniques for accurate lung cancer prediction

<https://inlibrary.uz/index.php/tajet/article/view/43510>

- Sweet M M R, Ahmed M P, Mozumder M A S, et al. Comparative analysis of machine learning techniques for accurate lung cancer prediction[J]. The American Journal of Engineering and Technology, 2024, 6(09): 92-103.

使用了包含有人口统计变量，生活方式因素，临床特征和环境暴露因素的数据集训练不同模型并进行评估。发现XGBoost的性能最好（准确性，敏感性，特异性，f1评分），LightGBM表现也很好，但略微逊色于前者。

同时还发现了肺癌的重要预测因素中包含吸烟史，家族史，和空气污染。

Optimized Deep Learning Model for Lung Cancer Prediction Using ANN Algorithm

<https://ieeexplore.ieee.org/abstract/document/9885607>

- Rawat D, Pawar L, Bathla G, et al. Optimized deep learning model for lung cancer prediction using ANN algorithm[C]//2022 3rd International Conference on Electronics and Sustainable Communication Systems (ICESC). IEEE, 2022: 889-894.

这项研究考虑了四种机器学习算法，如贝叶斯网、朴素贝叶斯、决策树、随机森林和一种深度学习算法人工神经网络，以预测早期肺癌。为此，测量不同算法的最新参数，并在肺癌数据集上进行评估。对所有五种算法的准确度作为主要参数进行评估和比较。实验结果显示，人工神经网络是最佳学习算法，准确率为92.23%。进一步评估了人工神经网络对一层、二层和三层隐藏层的准确性并进行了比较。单层人工神经网络在实验中具有最高的精度。

Predicting the future risk of lung cancer: development, and internal and external validation of the CanPredict (lung) model in 19.67 million people and evaluation of

model performance against seven other risk prediction models

[https://www.thelancet.com/journals/lanres/article/PIIS2213-2600\(23\)00050-4/fulltext](https://www.thelancet.com/journals/lanres/article/PIIS2213-2600(23)00050-4/fulltext)

- Liao W, Coupland C A C, Burchardt J, et al. Predicting the future risk of lung cancer: development, and internal and external validation of the CanPredict (lung) model in 19·67 million people and evaluation of model performance against seven other risk prediction models[J]. The Lancet Respiratory Medicine, 2023, 11(8): 685-697.

本研究开发和验证用于英国肺癌筛查的风险预测模型——CanPredict（肺）模型，并将该模型的性能与其他七种风险预测模型进行比较。

KNN and XGBoost Algorithms for Lung Cancer Prediction

[https://search.ebscohost.com/login.aspx?](https://search.ebscohost.com/login.aspx?direct=true&profile=ehost&scope=site&authtype=crawler&jrnl=28307275&AN=161832801&h=UMPK3T1G1fGsr3b9MHoTbJ7VmF8q3Ri7gnD4voUIhfOPKDGNtirv6uJ1iFEpLPG%2BTW0r8KTmZgTdfJ175ZwtQ%3D%3D&crl=c)

[direct=true&profile=ehost&scope=site&authtype=crawler&jrnl=28307275&AN=161832801&h=UMPK3T1G1fGsr3b9MHoTbJ7VmF8q3Ri7gnD4voUIhfOPKDGNtirv6uJ1iFEpLPG%2BTW0r8KTmZgTdfJ175ZwtQ%3D%3D&crl=c](https://search.ebscohost.com/login.aspx?direct=true&profile=ehost&scope=site&authtype=crawler&jrnl=28307275&AN=161832801&h=UMPK3T1G1fGsr3b9MHoTbJ7VmF8q3Ri7gnD4voUIhfOPKDGNtirv6uJ1iFEpLPG%2BTW0r8KTmZgTdfJ175ZwtQ%3D%3D&crl=c)

- Wayahdi M R, Ruziq F. KNN and XGBoost Algorithms for Lung Cancer Prediction[J]. Journal of Science Technology (JoSTec), 2022, 4(1).

这个研究使用了K-Nearest Neighbor和XGBoost模型用于癌症预测